

A screenshot from the game Minecraft showing a peaceful village scene. Several wooden houses with thatched roofs are built on a grassy hillside. In the background, a body of water is visible under a blue sky with scattered white clouds. The terrain is composed of green grass blocks and brown dirt blocks, with some small trees and flowers scattered around. The overall atmosphere is calm and scenic.

WHEN BLOCKS BUILD SOCIETIES

Multi-Agent Systems in Minecraft

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Sources: Logo from textcraft.net; video from teamcraft-bench.github.io/videos/main_web.mp4.

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Minecraft's procedural worlds and flexible mechanics make it an ideal platform for controlled and emergent agent behavior

- Balances computational feasibility with real-world complexity
- Supports a wide range of tasks like navigation, resource gathering, construction, and social interaction
- Enables observation of emergent behaviors beyond what's possible in more constrained virtual or robotic environments

Paper compares two state-of-the-art multi-agent systems to uncover principles for sustainable AI collaboration

- Research question: How do Project Sid's self-organizing social systems compare to TeamCraft's task-oriented coordination frameworks in achieving sustainable multi-agent collaboration?
- Understanding these trade-offs can inform the design of robust collaborative AI for real-world applications

Project Sid shows that AI agents can spontaneously develop social structures, roles, and cultures mirroring human societies

- Simulates 10–1000+ agents in Minecraft, using the PIANO architecture for real-time, multimodal interaction
- Agents develop specialized roles, form and adapt rules, and transmit culture and beliefs-without explicit programming
- Demonstrates adaptability, resilience, and the emergence of complex social contracts



Source: Video from [youtube.com/watch?v=2tbaCh3KI90](https://www.youtube.com/watch?v=2tbaCh3KI90).

TeamCraft excels at evaluating and training agents for efficient multi-modal collaboration on complex yet structured tasks

- Features 55,000+ task variants using visual and textual prompts, with expert demonstrations for imitation learning
- Focuses on explicit role assignment, task decomposition, and coordination protocols
- Highlights current limits in generalization: performance drops on novel goals, scenes, and team compositions

Project Sid prioritizes adaptability and sustainability while TeamCraft optimizes for efficiency in known contexts

Dimension	Project Sid	TeamCraft
Organizational Structure	Self-organizing, adaptive	Predefined, task-driven
Communication & Coordination	Emergent, flexible	Structured, efficient
Learning & Adaptability	Reinforcement and social learning, adapts to change	Imitation and fine-tuning, low on novel tasks
Sustainability & Scalability	Long-term resilience, 10–1000+ agents	Short-term efficiency, limited by training data

AI accelerated literature review, synthesis, and drafting, but required careful human oversight for accuracy and attribution

- LLMs summarized papers, identified research gaps, and generated draft sections
- All AI generated content was reviewed, verified, and refined by the author
- Transparency is maintained by providing complete documentation of the conversation history in the paper's appendix

Combining emergent and structured approaches can produce collaborative AI systems that are both efficient and adaptable

- Hybrid designs may use structured protocols for routine tasks and emergent organization for novel challenges
- Insights from Minecraft transfer to robotics, business, and digital governance
- Future research: hybrid models, long-term studies, and integration with human teams

When Blocks Build Societies: Multi-Agent Systems in Minecraft

With all references below, let's open the floor for questions, feedback, and ideas

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